

FMCG Data Consolidation & Analytics Platform

Project Documentation

1. Introduction

This project demonstrates an end-to-end Lakehouse data engineering pipeline for an FMCG acquisition scenario where Company A acquires Company B. The solution consolidates heterogeneous enterprise datasets using the Medallion Architecture (Bronze → Silver → Gold) and implements data ingestion, schema harmonization, data cleaning, analytical modelling, incremental loading, SQL analysis and Power BI reporting.

2. Project Objectives

- Build a Medallion Architecture.
- Consolidate Company A and Company B datasets.
- Perform schema harmonization and data quality improvements.
- Build Gold Layer analytical tables.
- Demonstrate Delta Lake MERGE based incremental loading.
- Automate execution using Databricks Workflows.
- Develop Power BI dashboards.

Technologies Used: Databricks, Pyspark, Delta Lake, SQL, Power BI, Python Faker, Git, GitHub

3. Project Scope & Implementation Decisions

The original architecture considered Amazon S3 as the storage layer. Since this implementation was developed using Databricks Free Edition, Databricks Workspace Volumes were used instead while preserving the Lakehouse architecture.

Implementation decisions:

- Amazon S3 → Workspace Volumes.
- Enterprise sources → Synthetic enterprise datasets (generated to simulate two independent ERP systems).
- Power BI → Gold CSV exports through Power Query.
- Workflow orchestration → Databricks Workflows demonstrated and ready for scheduling.

4. Dataset Generation

Synthetic datasets were created for Company A and Company B.

Company A Customers were generated using the Python Faker library.

All remaining datasets (Products, Stores, Orders for both companies and Company B Customers) were generated using AI-assisted (LLM) tools and validated.

Referential integrity was maintained throughout. Duplicate records, missing values, schema differences, inconsistent formatting, mixed customer IDs and negative quantities were intentionally introduced to simulate realistic enterprise data.

The following master and transactional datasets were created for Company A:

- customers_A.csv – Customer master data
- products_A.csv – Product master data
- stores_A.csv – Store master data
- orders_A.csv – Transactional order data

These datasets follow a standardized schema and include intentional data quality issues such as missing values, duplicate records, inconsistent formatting, and invalid entries to simulate real-world data.

Company B Datasets

A second set of datasets was generated for Company B using a slightly different schema to mimic data coming from another ERP system:

- customers_B.csv
- products_B.csv
- stores_B.csv
- orders_B.csv

The Company B datasets contain different column names for selected attributes while retaining some common fields, creating a realistic schema harmonization scenario during the Silver layer transformation.

Data Characteristics

The generated datasets include realistic FMCG business data with:

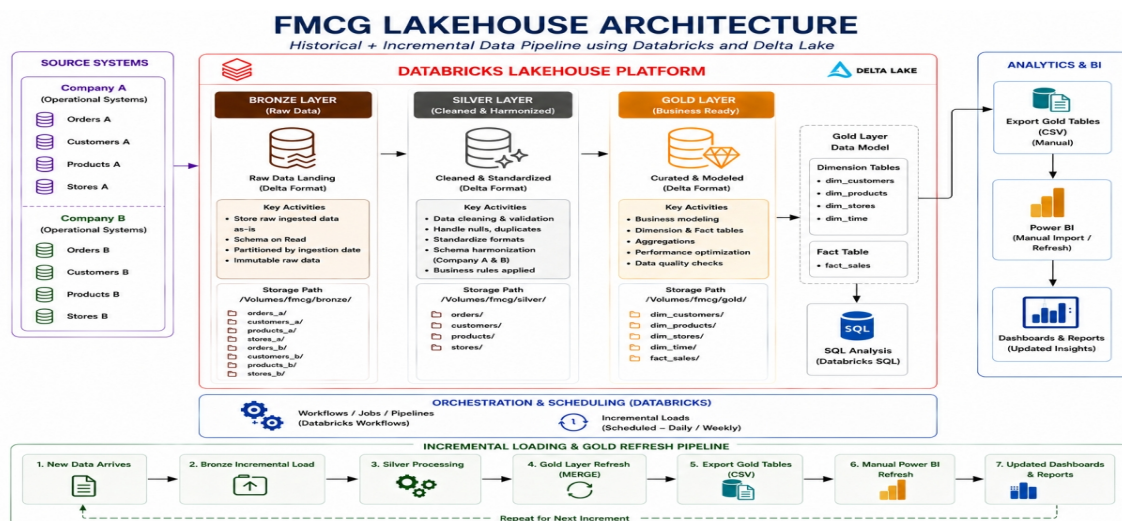
- Historical records from January 2023 to March 2026
- Referential integrity between master and transactional datasets
- Partial overlap of customers, products, and stores between Company A and Company B

- Intentionally introduced data quality issues, including:
 - Missing values
 - Duplicate records
 - Inconsistent text casing
 - Mixed formatting
 - Invalid quantity values
 - Slight differences in overlapping master records

5. Solution Architecture

Historical CSV Files → Bronze Layer → Silver Layer → Gold Layer → SQL Analysis & Power BI.

The solution follows the Medallion Architecture to progressively improve data quality across Bronze, Silver, and Gold layers. Historical datasets from Company A and Company B are ingested into the Bronze layer, transformed and harmonized in the Silver layer, and modeled into Star Schema tables in the Gold layer. These Gold tables support SQL analytics and Power BI dashboards, while incremental data is processed using Delta Lake MERGE operations orchestrated through Databricks Workflows.



6. Medallion Architecture

Note: Display statements were commented out before exporting the notebooks to reduce notebook size and improve GitHub readability. The transformation logic remains unchanged.

Bronze:

- Historical CSV ingestion
- Delta tables
- Raw data preservation

Silver:

Schema harmonization, duplicate removal, business-rule implementation, standardisation and consolidation.

Overall, the Silver layer includes:

Customers

- Removed duplicates
- Handled missing values
- Harmonized schema
- Unified Company A & B

Products

- Standardized categories (initcap)
- Standardized brands
- Handled null prices
- Harmonized schema
- Unified Company A & B

Stores

- Standardized city/state/region
- Handled missing manager
- Harmonized schema
- Unified Company A & B

Orders

- Removed duplicates
- Filled missing discounts
- Standardized payment modes

- Corrected negative quantities
- Harmonized schema
- Unified Company A & B

Gold:

- Built a Star Schema consisting of one fact table and multiple dimension tables.
- Created fact_sales,dim_customers,dim_products,dim_stores,dim_time.
- Calculated business metrics such as Sales Amount.

7. Incremental Processing & Workflow

Incremental CSV → Load → Silver MERGE → Gold Refresh .

Databricks Workflows automate execution using Incremental_Load followed by Gold_Layer_Update using the Gold_layer notebook.

8. SQL Analysis & Power BI

SQL notebooks provide revenue, customer, product and regional analysis using the gold data obtained right after historical load. The Power BI dashboard was created using the Gold Layer generated immediately after the historical load. Incremental loading demonstrates future updates separately.

9. Results

The project successfully demonstrates Medallion Architecture, schema harmonization, Delta Lake, incremental loading, workflow orchestration, SQL analytics and Power BI reporting.

Key Outcomes

- Successfully consolidated data from two independent companies.
- Harmonized different schemas into a unified model.
- Implemented historical and incremental batch processing.
- Built analytical Gold Layer using a Star Schema.
- Automated pipeline execution using Databricks Workflows.
- Enabled business reporting through SQL and Power BI.

10. Future Enhancements

Amazon S3 integration, Auto Loader, streaming pipelines, direct Power BI connectivity, CI/CD, data quality monitoring and Slowly Changing Dimensions.